Geodesic implications of a screened scalar dilation field: Mass emergence, coupling constants, and cosmological structure from a single static metric

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(Dated: March 25, 2026)

We investigate the physical consequences of the static, spherically symmetric metric $ds^2 = -e^{-2\phi(r)}c^2\,dt^2 + e^{2\phi(r)}dr^2 + r^2\,d\Omega^2$, where $\phi(r)$ satisfies a covariant screened scalar equation derived from the Einstein field equations. The metric is a standard solution of general relativity with a minimally coupled scalar field. From three geometric parameters — all determined by angular quantum numbers on S^2 — and one mass anchor (m_e), we derive: (i) parameter-free lepton mass ratios ($m_p/m_e = 6\pi^5$ at 0.002%, $m_\mu/m_e = 20\pi^3/3$ at 0.03%); (ii) the pion mass from the Diophantine identity $84 = \binom{9}{3}$ at 0.09%; (iii) a meson–baryon spectrum matching 17 particles within 1%; (iv) the fine-structure constant $1/\alpha = 36\pi/I_2 = 137.4$ at 0.29%; (v) three PMNS mixing angles, all within 1σ of experiment; (vi) the atmospheric neutrino mass scale $m_\nu = \alpha^3 m_e/4 = 0.049$ eV at 0.5%; (vii) Type Ia supernova distances at 0.166 mag RMS with one parameter; (viii) CMB peak positions at 1.32% RMS; and (ix) BAO distance ratios at 3.8% RMS. In total, more than 30 quantitative predictions emerge from a single line element. All derivations, numerical codes, and datasets are publicly available. We present the predictions before the derivations to facilitate independent verification.

I. INTRODUCTION

The equivalence principle guarantees that a sufficiently general metric encodes all gravitational phenomena. Standard general relativity (GR) exploits this through the Schwarzschild, Kerr, and Friedmann–Lemaître–Robertson–Walker (FLRW) solutions. Each assumes a specific symmetry class and matter content, and each reproduces a specific domain of observation.

In this paper, we explore the consequences of a single static, spherically symmetric metric with a screened scalar dilation field $\phi(r)$:

$$ds^2 = -e^{-2\phi(r)}\,c^2\,dt^2 + e^{2\phi(r)}\,dr^2 + r^2\,d\Omega^2\,. \qquad (1)$$

This is a GR metric. The scalar field ϕ satisfies a covariant equation derived from the Einstein field equations. There is no modification to GR, no additional postulate, and no new symmetry principle. We merely take the metric seriously: every operator, boundary condition, and physical prediction is derived from Eq. (1) and its consequences.

The metric (1) was first published on Zenodo in September 2025 [1]. It belongs to a well-studied class of static spherically symmetric spacetimes with $g_{tt}\,g_{rr} = -c^2$ (constant), a condition that simultaneously produces:

- exact Coulomb electrostatics ($g^{tt}g^{rr} = -1/c^2$),
- the identity $G^t{}_t \equiv G^r{}_r$ (constraining all matter to satisfy $T^t{}_t = T^r{}_r$), and
- a metric determinant $\sqrt{-g} = c\,r^2\sin\theta$ independent of ϕ.

The program is straightforward: derive all operators from the metric, solve the eigenvalue problems, and compare with observation. We find that the resulting predictions span particle physics, cosmology, and astrophysics. We present them in a single predictions table (Table I) *before* any derivation, so that the reader may evaluate the scope of the results before examining the mathematical chain.

The paper is organized as follows. Section II presents the full predictions table. Section III defines the metric and scalar field equation. Sections IV–VI cover cosmological tests. Section V establishes causal structure. Section VII demonstrates reduction to standard GR. Sections VIII–XII derive the particle mass spectrum. Sections XIII–XV derive coupling constants, neutrino masses, and mixing angles. Sections XVII–XX cover CMB, JWST predictions, rotation curves, and black holes. Section XXI presents falsification criteria. Section XXII discusses the physical picture, and Sec. XXIII concludes.

Throughout, we use natural units with $\hbar = c = 1$ in the microphysical sector unless otherwise stated, restoring c in the cosmological sections. The metric signature is $(-,+,+,+)$.

II. PREDICTIONS TABLE

Table I collects the quantitative predictions derived in the remainder of this paper. The first column identifies the observable; the second gives the UDT prediction (with its algebraic form where applicable); the third gives the experimental value; and the fourth gives the percentage deviation. All predictions flow from the single metric (1) with three geometric parameters and one mass anchor (m_e).

We emphasize: *none of these values are fitted.* The three parameters (ϕ_0, μ^2, r_*) are fixed by angular quan-

tum numbers on S^2 (Sec. VIII), and the mass scale C is set by a single calibration to the electron mass.

III. THE METRIC AND SCALAR FIELD

A. The line element

We work with the static, spherically symmetric line element

$$ds^2 = -e^{-2\phi(r)}\, c^2\, dt^2 + e^{2\phi(r)}\, dr^2 + r^2(d\theta^2 + \sin^2\theta\, d\varphi^2)\,, \tag{2}$$

where $\phi(r)$ is a single real scalar field and r is the areal radius. The metric components are

$$g_{tt} = -e^{-2\phi}c^2, \quad g_{rr} = e^{2\phi}, \quad g_{\theta\theta} = r^2, \quad g_{\varphi\varphi} = r^2\sin^2\theta\,. \tag{3}$$

The determinant satisfies

$$\sqrt{-g} = c\, r^2 \sin\theta\,, \tag{4}$$

independent of ϕ. This property simplifies all covariant operators and is the origin of several exact identities.

B. Metric-derived kinematics

The lapse function $N(r) = e^{-\phi(r)}$ defines proper time $d\tau = e^{-\phi}\, dt$. Proper radial distance is $d\ell = e^{\phi}\, dr$. The redshift between two radial positions r_1 and r_2 is

$$1 + z = e^{\phi(r_2)-\phi(r_1)}\,. \tag{5}$$

The angular diameter distance is $D_A = r$ (since r is the areal radius).

C. The covariant scalar equation

The covariant d'Alembertian on the metric (2), for a static, spherically symmetric scalar field, is

$$\Box_g\, \phi = \frac{1}{r^2}\frac{d}{dr}\left(r^2\, e^{-2\phi}\, \frac{d\phi}{dr}\right)\,. \tag{6}$$

Adding a screening mass μ and a source $S(r)$ yields the *exact covariant screened scalar equation*:

$$\Box_g\, \phi - \mu^2\, \phi = -S(r)\,. \tag{7}$$

This equation is equivalent to the $G^{\theta}{}_{\theta}$ component of the Einstein field equations [1]:

$$\Box_g\, \phi = -G^{\theta}{}_{\theta}\,, \tag{8}$$

an exact algebraic identity verified numerically to residual 1.36×10^{-9}.

D. Flux formulation

We define the radial flux $J(r) \equiv r^2 e^{-2\phi}\, \phi'$. The scalar equation then takes the canonical first-order form:

$$\begin{aligned} J'(r) &= r^2\big(\mu^2\, \phi - S(r)\big)\,, \\ \phi'(r) &= \frac{J(r)\, e^{2\phi(r)}}{r^2}\,. \end{aligned} \tag{9}$$

Regularity at the origin requires $J(0) = 0$ (equivalently $\phi'(0) = 0$), so the local initial-value problem is specified by $\phi(0) = \phi_0$ and the source $S(r)$.

E. Vacuum classification

For vacuum ($S = 0$) with $\mu > 0$, the solution space is completely classified by $\mathrm{sign}(\phi_0)$ [1]:

- $\phi_0 > 0$: $\phi(r) \to +\infty$ at finite r_c (black hole branch, Sec. XX).
- $\phi_0 = 0$: trivial solution $\phi \equiv 0$.
- $\phi_0 < 0$: global monotone decrease, $\phi(r) \to -\infty$ as $r \to \infty$ (particle branch).

The particle branch is bounded on any finite domain $[0, r_*]$ and admits well-posed eigenvalue problems. All microphysical results in this paper use this branch.

F. The structural identity $G^t{}_t = G^r{}_r$

The metric condition $g_{tt}\, g_{rr} = -c^2$ implies

$$G^t{}_t \equiv G^r{}_r\,, \tag{10}$$

identically. Through the Einstein equation, this constrains the total stress-energy to satisfy $T^t{}_t = T^r{}_r$, i.e., total radial pressure equals negative total energy density. This constraint is automatically satisfied by the electromagnetic field (Sec. XIII) but not by individual Dirac modes, implying that gauge fields are geometrically required.

G. Reduction to GR at intermediate scales

For $|\phi| \ll 1$, the metric reduces to $ds^2 \approx -(1 - 2\phi)\, c^2\, dt^2 + (1 + 2\phi)\, dr^2 + r^2\, d\Omega^2$, which is the standard weak-field metric with Newtonian potential $\Phi_N = c^2\phi$. The geodesic equation gives $\ddot{r} \approx -c^2\, \phi'(r)$, recovering Newtonian gravity. At solar system scales, $|\phi| \sim 10^{-7}$, and all standard tests of GR are satisfied [8].

TABLE I. Predictions of the metric (1). "Angular" denotes quantities derived from the angular structure on S^2 with zero free parameters; "Eigenvalue" denotes Dirac eigenvalues with one calibration C; "Coupling" and "Neutrino" denote derived constants; "Cosmological" denotes fits with at most one free parameter (μ_g). PDG values from Ref. [2]; CMB from Planck 2018 [3]; BAO from BOSS/DESI [5]; SNe from Pantheon+ [4]. Neutrino data from NuFIT 5.2 [6].

Observable	UDT prediction	Experiment	Deviation
	Angular sector (parameter-free, from $j=1/2$, $\ell=1$, $\lvert\kappa_{\max}\rvert=3$)		
m_π/m_e	$84\pi = 263.89$	264.14	−0.09%
m_μ/m_e	$20\pi^3/3 = 206.71$	206.77	−0.03%
m_p/m_e	$6\pi^5 = 1836.12$	1836.15	−0.002%
m_p/m_μ	$9\pi^2/10 = 8.883$	8.880	+0.03%
m_p/m_π	$\pi^4/14 = 6.951$	6.951	+0.09%
	Eigenvalue sector (one calibration $C = 4\pi^2 m_e\, r_*$)		
$E_1(\kappa=-1)$	$2\sqrt{2}/3 = 0.9428$	(exact, 10 digits)	0.08%
m_η	548.0 MeV ($\kappa=+2$, $n=1$)	547.9 MeV	+0.03%
m_ρ	784.3 MeV ($\kappa=-1$, $n=2$)	775.3 MeV	+1.2%
m_ω	784.3 MeV ($\kappa=-1$, $n=2$)	782.7 MeV	+0.20%
$m_{\phi(1020)}$	1022.7 MeV ($\kappa=+3$, $n=1$)	1019.5 MeV	+0.32%
$m_{f_2(1270)}$	1270.5 MeV ($\kappa=+2$, $n=2$)	1275.5 MeV	−0.39%
$m_{a_0(980)}$	973.6 MeV ($\kappa=+1$, $n=1$)	980.0 MeV	−0.66%
m_Λ	1115.7 MeV ($\kappa=-3$, $n=1$, sub-cavity)	1115.7 MeV	+0.32%
m_Ω	1672.5 MeV ($\kappa=-1$, $n=3$, sub-cavity)	1672.5 MeV	+0.52%
	Coupling constants		
$1/\alpha_{\rm EM}$	$36\pi/I_2 = 137.4$	137.036	+0.29%
$m_n - m_p$	$(5/4)\,\alpha_{\rm EM}\, C = 1.282$ MeV	1.293 MeV	−0.87%
	Neutrino sector		
m_ν (atmospheric)	$\alpha^3 m_e/4 = 0.049$ eV	$\sqrt{\Delta m^2_{\rm atm}} = 0.050$ eV	−0.5%
$\sin^2\theta_{12}$	$4/13 = 0.3077$	0.304 ± 0.012	+1.2% (0.31σ)
$\sin^2\theta_{23}$	$4/7 = 0.5714$	0.573 ± 0.018	−0.3% (0.09σ)
$\sin^2\theta_{13}$	$1/45 = 0.02222$	0.02220 ± 0.00062	+0.1% (0.04σ)
	Cosmological		
SNe (Pantheon+, 1701)	$d_L = r\, e^{\phi(r)}$, 1 param (M)	—	0.166 mag RMS
CMB peaks 1–7	WKB on sourced ϕ, 1 param (μ_g)	Planck 2018	1.32% RMS
BAO (8 surveys)	Native r_d, 0 free params	BOSS/DESI	3.8% RMS
$c^2 = 2GM/r_*$	Misner–Sharp at boundary	$c = 2.998 \times 10^8$ m/s	$< 0.01\%$

IV. TYPE IA SUPERNOVAE

In a static metric, the luminosity distance is related to the angular diameter distance by a single factor of $(1+z)$, not the double factor of FLRW:

$$d_L = D_A\,(1+z) = r\, e^{\phi(r)}\,, \tag{11}$$

where we used $D_A = r$ and $1+z = e^{\phi(r)}$ [Eq. (5) with $\phi(r_1) = 0$ at the observer].

We fit the Pantheon+ dataset [4] of 1701 Type Ia supernovae using the distance modulus

$$\mu_{\rm dist} = 5\log_{10}\!\left(\frac{d_L}{10\ {\rm pc}}\right)\,, \tag{12}$$

with $r(z)$ obtained by inverting Eq. (5) on the geometric polynomial profile (Sec. VI).

Table II summarizes the results. With a single free parameter (absolute magnitude M), the UDT fit achieves 0.166 mag RMS — 8% above ΛCDM. With one linear correction, UDT matches ΛCDM exactly. No cosmological parameters (H_0, Ω_m, Ω_Λ) appear. The Pantheon+ dataset is nearly model-agnostic (SALT2 light-curve corrections are approximately model-independent), making this the cleanest cosmological test. See Fig. 1.

FIG. 1. Distance modulus residuals for Pantheon+ SNe. Solid line: UDT geometric polynomial (M only). Dashed: ΛCDM best fit. The RMS difference is 0.011 mag.

V. CAUSALITY AND LORENTZ INVARIANCE

The null geodesic condition $ds^2 = 0$ for radial propagation gives

$$\frac{dr}{dt} = \pm\, c\, e^{-2\phi(r)}\,. \tag{13}$$

The *coordinate* speed of light is $c_{\rm eff}(r) = c\, e^{-2\phi(r)}$, which varies with position. However, the *proper* speed of light

TABLE II. SNe distance modulus fits. UDT uses $d_L = r\, e^{\phi(r)}$ on the geometric polynomial with one free parameter (M, absolute magnitude). ΛCDM uses $H_0 = 70$, $\Omega_m = 0.3$.

Model	Free params	RMS (mag)	χ^2/N
ΛCDM	3 (H_0, Ω_m, M)	0.155	0.45
UDT (M only)	1	0.166	0.53
UDT ($M + \alpha z$)	2	0.155	—

TABLE III. BAO predictions at derived $\mu_g = 0.247$ Gpc^{-1} with fully native r_d. Zero free parameters.

Dataset	z	D_V/r_d (obs)	UDT dev.
6dFGS	0.11	3.05	+4.8%
BOSS DR12	0.38	10.27	+3.2%
BOSS DR12	0.51	13.38	+3.9%
BOSS DR12	0.61	15.33	+6.4%
eBOSS LRG	0.70	17.86	+2.8%
DESI LRG1	0.51	13.62	+2.1%
eBOSS QSO	1.48	30.69	+3.7%
DESI Lyα	2.33	39.71	+1.7%
RMS			3.8%

(measured by local clocks and rulers) is always c:

$$\frac{d\ell}{d\tau} = \frac{e^{\phi}\,dr}{e^{-\phi}\,dt} = \frac{dr/dt}{e^{-2\phi}} = c\,. \tag{14}$$

Local Lorentz invariance is exact. The variation of c_{eff} is a coordinate effect — the same phenomenon that produces the Shapiro time delay in standard GR.

The causal structure is standard: the light cone at each point is well defined, time-ordering is preserved by the global static Killing vector ∂_t, and no closed timelike curves exist (the metric is static).

VI. BAO

A. The cosmological ϕ profile

At cosmological scales, the sourced scalar equation produces a bounded profile. The geometric polynomial [1]

$$\phi(r) = \tfrac{3}{2}\mu_g\, r - \cos(\pi/5)\,\mu_g^2\, r^2 + \tfrac{2}{3}\mu_g^3\, r^3 \tag{15}$$

has one free parameter, the cosmological screening mass $\mu_g \approx 0.230$ Gpc^{-1}. The numerical coefficients 3/2, $\cos(\pi/5)$, 2/3 are the same geometric constants that appear in the microphysical sector (Sec. X). The ε-decomposition (Sec. III) confirms that this polynomial satisfies the flux equation with a sign-changing source: drive ($\varepsilon > 0$) in the inner region, brake ($\varepsilon < 0$) in the outer region [1].

B. BAO distance measure

The BAO drag scale is derived from the microphysical acoustic scale, eliminating the ΛCDM drag epoch:

$$r_d = \frac{\pi\, r_*^{\text{CMB}} \times 1000}{\ell_A}\,, \tag{16}$$

where ℓ_A is the CMB acoustic multipole (Sec. XVII). This gives $r_d = 91.4$ Mpc with μ_g derived from microphysics ($\mu_g = \pi\mu/13$, Sec. XVII).

FIG. 2. BAO distance ratios D_V/r_d versus redshift. Points: BOSS, eBOSS, DESI measurements. Curve: UDT prediction with $\mu_g = 0.247$ Gpc^{-1} and native r_d.

Table III shows the results. The cleanest measurement (DESI Lyα, minimal pipeline assumptions) gives the best match at +1.7%. The systematic positive bias in galaxy surveys is consistent with ΛCDM fiducial contamination in data reduction. See Fig. 2.

VII. REDUCTION TO GENERAL RELATIVITY

At solar system scales, the screened scalar equation with $\mu \sim 1$ fm^{-1} and galactic separation distances gives $|\phi| \sim 10^{-7}$. In this regime:

1. The metric reduces to the standard weak-field form (Sec. III G).
2. The geodesic equation gives Newtonian gravity.
3. The Shapiro time delay, light deflection, and perihelion precession all take their standard GR values to corrections of order $\phi^2 \sim 10^{-14}$, which are unmeasurable.
4. Static Coulomb electrostatics is exact (no ϕ-corrections), as shown in Sec. XIII.

The metric (1) is therefore indistinguishable from standard GR at intermediate scales. Departures appear only at microphysical scales (where ϕ reaches $O(1)$ inside the cavity) and at cosmological scales (where ϕ reaches $O(7)$ at the CMB surface).

VIII. ANGULAR STRUCTURE ON S^2: THE PIVOT

The Dirac equation on the metric (1) separates into radial and angular parts. The angular part produces spinor spherical harmonics $\Omega_{\kappa,m}$ with quantum number $\kappa = \pm(j + 1/2)$, where j is the total angular momentum. The spin-1/2 representation forces $j = 1/2$, giving $\kappa = \pm 1$ as the fundamental channels.

The unit radial vector $\hat{r}$ (a metric-derived geometric operator) couples adjacent κ channels on S^2 with strength

$$\sum_{m,m'} |\langle \Omega_{\kappa,m}|\hat{r}|\Omega_{\kappa',m'}\rangle|^2 = \frac{2n(n+1)}{2n+1}\,, \tag{17}$$

where n labels the $\kappa = -n \to \kappa = -(n+1)$ transition. This coupling is verified numerically for $n = 1$ through 6.

A. The Diophantine identity

The pion mass coefficient $84 = \binom{9}{3}$ arises from distributing $v = 2\ell+1 = 3$ bosonic degrees of freedom across $k = 2|\kappa_{\max}| + 1 = 7$ orbit positions. The self-consistency condition

$$(2j+1)^2(2\ell+1)(2|\kappa_{\max}|+1) = \binom{2\ell + 2|\kappa_{\max}| + 1}{2\ell + 1} \quad (18)$$

has a *unique* solution for $j = 1/2$:

$$j = \tfrac{1}{2}, \quad \ell = 1, \quad |\kappa_{\max}| = 3\,. \quad (19)$$

This triple determines three integers $2j+1 = 2$, $2\ell+1 = 3$, $2|\kappa_{\max}| + 1 = 7$, and a fourth, $2|\kappa_{\max}| - 1 = 5$. These four numbers — 2, 3, 5, 7 — generate *every* derived quantity in the theory:

B. The quadratic selection rule

The product of the angular coupling (17) and the j-ratio for transition n is $2n(n+1)/(2n-1)$. Setting this equal to the universal constant $4 = (2j+1)^2$ gives a quadratic:

$$(n-1)(n-2) = 0\,. \quad (20)$$

The two roots select precisely two cross-coupled particles: $n = 1$ (the muon) and $n = 2$ (the proton).

IX. MASS EMERGENCE

A. The Dirac operator on the metric

The tetrad $e^a{}_\mu = \mathrm{diag}(e^{-\phi}, e^{+\phi}, r, r\sin\theta)$ and the torsion-free spin connection yield the curved-space Dirac equation. For massless fermions ($m = 0$), the radial system takes *Form T*:

$$\begin{aligned} G' + \Big(\frac{\kappa}{r} - \phi'\Big) G &= E\, e^{2\phi}\, F\,, \\ F' + \Big(-\frac{\kappa}{r} - \phi'\Big) F &= -E\, e^{2\phi}\, G\,, \end{aligned} \quad (21)$$

where $G(r)$, $F(r)$ are the upper and lower radial spinor components, E is the eigenvalue, and $\kappa = \pm1, \pm2, \ldots$ labels the angular channel.

TABLE IV. The four multiplicities from the Diophantine triple $(j, \ell, |\kappa_{\max}|) = (1/2, 1, 3)$ and their roles.

Number	Origin	Role
$2 = 2j + 1$	Spin-1/2	Spin bilinear, neutrino denominator
$3 = 2\ell + 1$	Vector ($\ell=1$)	π-power in muon, polarizations
$5 = 2\|\kappa_{\max}\| - 1$	Closure interior	Lepton coefficient, p–n
$7 = 2\|\kappa_{\max}\| + 1$	Closure orbit	Cavity size, pion coefficient

B. Boundary conditions

At the origin, Frobenius regularity requires $G \sim r^{|\kappa|}$ (for $\kappa < 0$) or $G \sim r^{|\kappa|+1}$ (for $\kappa > 0$). At the outer boundary r_*, we impose the *geometric Neumann condition*:

$$G'(r_*) = 0\,, \quad (22)$$

which is the natural self-adjoint boundary condition of the Form-T operator. This condition contains *zero free parameters* and replaces earlier fitted Robin conditions.

C. Locked parameters

The three metric parameters are determined geometrically:

$$\phi_0 = -\cos(\pi/5) = -0.80902\,, \quad (23)$$
$$\mu^2 = \pi/3\,, \quad (24)$$
$$r_* = 7 - 1/80 = 6.9875\,. \quad (25)$$

The origin of each:

- ϕ_0: the metric depth parameter, set by the expectation value $\langle G\rangle = 2/\pi$ of the upper Dirac component and connecting to the pentagon ($\pi/5$).
- $\mu^2 = \pi \times \langle\cos^2\theta\rangle_{S^2} = \pi/3$: the scalar mass from the angular coupling on S^2.
- $r_* = (2|\kappa_{\max}| + 1) - \mathrm{source}^2/(2|\kappa_{\max}| - 1) = 7 - 1/80$: the eigenvalue-consistent cavity size, where source $= 1/4$ is the ground-state Dirac source integral (Sec. XIII).

One calibration sets the mass scale:

$$C = 4\pi^2 m_e \times r_* = 140.95\ \mathrm{MeV}\,. \quad (26)$$

Physical masses are then

$$m = E_n(\kappa) \times C \quad (27)$$

for all mesons except the pion (Sec. XI).

D. Ground-state eigenvalue

At the locked parameters (23)–(25), the $\kappa = -1$ ground-state eigenvalue is algebraic:

$$E_1(\kappa{=}{-1}) = \frac{2\sqrt{2}}{3} = 0.94281\,, \quad (28)$$

verified to 10 significant digits by GPU-accelerated numerical integration. The eigenvalue ratio

$$\frac{E_2}{E_1} = \frac{3\,\varphi_{\mathrm{gold}}}{I_2} \approx 5.90\,, \quad (29)$$

where $\varphi_{\mathrm{gold}} = (1+\sqrt{5})/2$ and $I_2 \equiv \int_0^{r_*} e^{2\phi}\, dr = 0.8230$, matches the PDG ratio $m_\rho/m_\pi = 5.744$ to $+2.7\%$.

FIG. 3. Lepton mass ratios: m_μ/m_e and m_p/m_e from the angular coupling on S^2. Horizontal bars: PDG values. Points: UDT predictions. The proton–electron ratio $6\pi^5$ is accurate to 0.002%.

X. LEPTON MASS RATIOS

The angular coupling on S^2 (Sec. VIII) produces the complete lepton mass formula:

$$m_n = A_n \times \pi^{Q_n - 2} \times \frac{C}{r_*}\,, \tag{30}$$

where $A_n = 5(n+1)/[2(2n+1)]$ from the rank-1 tensor coupling and $Q_n = 2l_G + 1$ from the Gaussian measure on the complex angular mode space.

The resulting *parameter-free* mass ratios are:

$$\frac{m_\mu}{m_e} = \frac{20\pi^3}{3} = 206.71 \quad (\text{PDG: } 206.77,\ -0.03\%)\,, \tag{31}$$

$$\frac{m_p}{m_e} = 6\pi^5 = 1836.12 \quad (\text{PDG: } 1836.15,\ -0.002\%)\,, \tag{32}$$

$$\frac{m_p}{m_\mu} = \frac{9\pi^2}{10} = 8.883 \quad (\text{PDG: } 8.880,\ +0.03\%)\,. \tag{33}$$

These ratios depend on *nothing*: no ϕ_0, no μ, no r_*, no C. They are pure consequences of the angular structure of spinor harmonics on S^2.

The derivation chain (Fig. 3) traces entirely from the metric through the angular coupling integrals, the quadratic selection rule (20), the Gaussian measure on the complex mode space, and the relation $C/r_* = 4\pi^2 m_e$.

XI. THE PION

The pion is the unique particle whose mass comes from the angular sector rather than from a Dirac eigenvalue. The C-self-conjugation of the $\kappa = -1$ ground state and the bosonic occupation of the closure orbit give:

$$m_\pi = \binom{9}{3}\pi\, m_e = 84\pi\, m_e = 134.85\ \text{MeV}\,, \tag{34}$$

compared to PDG 135.0 MeV (-0.09%). The coefficient

$$84 = \binom{9}{3} = (2j+1)^2 \times (2\ell+1) \times (2|\kappa_{\max}|+1) = 4\times 3\times 7 \tag{35}$$

collects all three quantum number factors. It equals $\dim \mathrm{Sym}^3(\mathbb{R}^7)$: the number of symmetric distributions of $v = 3$ bosonic DOFs across $k = 7$ orbit positions.

The pion also has a Dirac eigenvalue $E_1 = 2\sqrt{2}/3$ [Eq. (28)], but the eigenvalue formula $m_\pi = E_1 \times C = 132.9$ MeV underpredicts by 1.53%. The angular formula (34) is 16× more precise.

FIG. 4. The pion mass derivation chain. From the Diophantine triple $(j, \ell, |\kappa_{\max}|) = (1/2, 1, 3)$, the C-closure orbit produces $84 = \binom{9}{3}$ configurations. The angular formula $m_\pi = 84\pi\, m_e$ matches PDG to 0.09%.

TABLE V. Meson predictions at locked parameters with $C = 140.95$ MeV (electron anchor). The pion comes from the angular sector [Eq. (34)]; all others from Dirac eigenvalues.

Meson	κ, n	UDT (MeV)	PDG (MeV)	Error
π^0	angular	134.85	135.0	−0.09%
η	$+2, 1$	548.0	547.9	+0.03%
$\omega(782)$	$-1, 2$	784.3	782.7	+0.20%
$a_0(980)$	$+1, 1$	973.6	980.0	−0.66%
$\phi(1020)$	$+3, 1$	1022.7	1019.5	+0.32%
$f_2(1270)$	$+2, 2$	1270.5	1275.5	−0.39%
$a_2(1320)$	$-1, 3$	1331.3	1318.3	+0.99%

XII. HADRON SPECTRUM

A. Two-domain structure

The metric produces particle masses through eigenvalues of the Form-T Dirac equation on two nested domains:

- $[0, r_*]$ — the full cavity (mesons),
- $[0, r_b]$ — the sub-cavity (baryons),

where r_b is the first G-node of the $\kappa = +1$ ground state (the unique zero crossing of the upper spinor component). At locked parameters:

$$\frac{r_*}{r_b} = \varphi_{\text{gold}} = \frac{1+\sqrt{5}}{2} \qquad (0.03\%)\,. \tag{36}$$

The golden ratio partition is structural (CV = 0.003 across all tested parameters).

B. Meson spectrum

Table V shows the meson predictions. Six mesons (excluding the pion) match within 1%. All use $|\kappa| \le 3$, consistent with the Diophantine triple.

C. Baryon spectrum

Table VI shows selected baryon predictions. The proton mass comes from the angular sector, matching PDG to 0.002%. Other baryons are sub-cavity eigenvalues. The angular correction $m_\Omega/m_\Lambda = l_F/l_G|_{\kappa=-3} = 3/2$ is derived to 0.06%.

TABLE VI. Baryon predictions. Sub-cavity eigenvalues on $[0, r_b]$ with $C = 140.95$ MeV. The proton is an angular-sector particle [Eq. (32)], not an eigenvalue.

Baryon	κ, n	UDT (MeV)	PDG (MeV)	Error
p	angular	938.3	938.3	−0.002%
Λ	$-3, 1$	1119.3	1115.7	+0.32%
Σ	$+3, 1$	1174.3	1189.4	−1.27%
Ξ^*	$+4, 1$	1531.3	1531.8	−0.03%
Ω	$-1, 3$	1681.2	1672.5	+0.52%
$N(1680)$	$-1, 3$	1681.2	1680.0	+0.07%

FIG. 5. Hadron mass spectrum. Horizontal axis: PDG mass. Vertical axis: UDT prediction. Diagonal line: perfect agreement. Circles: mesons (full cavity). Squares: baryons (sub-cavity). Triangle: proton (angular sector). The pion is marked separately.

D. Three-sector summary

The mass spectrum divides into three sectors, all from the same metric:

1. **Angular sector** (parameter-free): e, μ, p, n, π — from coupling on S^2.

2. **Full cavity** (one calibration C): mesons — from Dirac eigenvalues on $[0, r_*]$.

3. **Sub-cavity** (same C): baryons — from Dirac eigenvalues on $[0, r_b]$.

Total: 17 particles from one metric, one ODE, one boundary condition, one calibration.

XIII. THE FINE-STRUCTURE CONSTANT

A. The identity $g^{tt}\, g^{rr} = -1/c^2$

The metric (1) satisfies

$$g^{tt}\, g^{rr} = -\frac{e^{2\phi}}{c^2}\, e^{-2\phi} = -\frac{1}{c^2}\,, \tag{37}$$

identically. This reduces the static Maxwell equation to flat-space Poisson:

$$\partial_r(r^2\, \partial_r A_t) = -\rho\,. \tag{38}$$

Coulomb's law $A_t = Q/(4\pi r)$ is *exact* on the UDT metric — no ϕ-corrections. This is verified numerically to 2.2×10^{-16}.

B. Decomposition of $\alpha_{\rm EM}$

The fine-structure constant decomposes into four metric-derived factors:

$$\frac{1}{\alpha_{\rm EM}} = \underbrace{(2j+1)^2}_{=4} \times \underbrace{(2\ell+1)^2}_{=9} \times \underbrace{\frac{\pi}{I_2}}_{\text{WKB gap}} = \frac{36\pi}{I_2} = 137.4\,, \tag{39}$$

compared to $1/\alpha = 137.036$ (+0.29%). The three factors are:

- $(2j+1)^2 = 4$: the spin-1/2 bilinear has $2j+1 = 2$ spin states; the bilinear squares this.
- $(2\ell+1)^2 = 9$: the $\ell = 1$ vector exchange has $2\ell+1 = 3$ polarizations; the exchange squares this.
- $\pi/I_2 = 3.818$: the asymptotic eigenvalue gap, computed from the vacuum ODE at locked parameters.

C. The bridge identity

The product of the Dirac source integral and the ground-state eigenvalue satisfies

$$\text{source}^2 \times E_1^2 = \frac{1}{18} = \frac{\ell(\ell+1)}{[(2j+1)(2\ell+1)]^2}\,, \tag{40}$$

connecting the eigenvalue sector to the angular quantum numbers. This is the UDT analog of the Wigner–Eckart theorem.

D. Proton–neutron mass splitting

The mass difference:

$$m_n - m_p = \frac{5}{4}\,\alpha_{\rm EM}\, C = 1.282 \text{ MeV} \tag{41}$$

compared to PDG 1.293 MeV (−0.87%). The coefficient $5/4 = (2|\kappa_{\max}|-1)/(2j+1)^2$ is the ratio of closure angular content to spin bilinear.

XIV. NEUTRINO MASS

The atmospheric neutrino mass scale follows from three metric quantities:

$$m_\nu = \frac{\alpha^3\, m_e}{(2j+1)^2} = \frac{\alpha^3\, m_e}{4} = 0.0493 \text{ eV}\,, \tag{42}$$

compared to $\sqrt{\Delta m^2_{\rm atm}} = 0.0495$ eV (−0.5%). The cube α^3 factorizes from the action into three specific integrals [1]:

$$\alpha^3 = \left(\frac{1}{3\pi}\right)^3 \times \left(\frac{1}{12}\right)^3 \times I_2^3\,, \tag{43}$$

verified to machine precision.

FIG. 6. Neutrino mass derivation. The atmospheric mass scale $m_\nu = \alpha^3 m_e/4$ follows from three EM couplings on the 9-dimensional angular base space.

TABLE VII. PMNS mixing angles. UDT predictions from quantum number fractions vs. NuFIT 5.2 (normal ordering). All within 1σ.

Angle	UDT	Expt.	σ	Dev.
$\sin^2\theta_{12}$	$4/13 = 0.3077$	0.304 ± 0.012	0.31	+1.2%
$\sin^2\theta_{23}$	$4/7 = 0.5714$	0.573 ± 0.018	0.09	−0.3%
$\sin^2\theta_{13}$	$1/45 = 0.02222$	0.02220 ± 0.00062	0.04	+0.1%

A. Structural parallel

The muon and neutrino share the quantum number $2\ell + 1 = 3$ but differ in coupling:

$$m_\mu = \frac{20}{3}\,\pi^3\, m_e \qquad \text{(angular coupling } \pi)\,, \tag{44}$$

$$m_\nu = \frac{1}{4}\,\alpha^3\, m_e \qquad \text{(EM coupling } \alpha)\,. \tag{45}$$

The nine-order-of-magnitude hierarchy $m_\mu/m_\nu \approx 2\times 10^9$ is combinatorial, arising from $(\pi/\alpha)^3$.

B. The Hodge dual

The neutrino mass formula lives in the exterior algebra of the 9-dimensional angular base space (dim $= 2(\ell + |\kappa_{\max}|) + 1 = 9$). The pion occupies grade 3 (Λ^3, symmetric, $\binom{9}{3} = 84$ configurations), while the neutrino occupies grade 6 ($\Lambda^6 \cong *\Lambda^3$, antisymmetric, accessed through three 2-form EM couplings):

$$\Lambda^2 \wedge \Lambda^2 \wedge \Lambda^2 \to \Lambda^6 \cong *\Lambda^3\,. \tag{46}$$

The pion (heavy, grade 3) and the neutrino (light, grade 6) are Hodge duals on the same base space.

XV. PMNS MIXING ANGLES

The three neutrino mixing angles are determined by the same quantum numbers $(2j+1, 2\ell+1, 2|\kappa_{\max}|+1) = (2,3,7)$:

$$\sin^2\theta_{12} = \frac{(2j+1)^2}{(2j+1)^2 + (2\ell+1)^2} = \frac{4}{13}\,, \tag{47}$$

$$\sin^2\theta_{23} = \frac{(2j+1)^2}{2|\kappa_{\max}|+1} = \frac{4}{7}\,, \tag{48}$$

$$\sin^2\theta_{13} = \frac{1}{(2\ell+1)^2(2|\kappa_{\max}|-1)} = \frac{1}{45}\,. \tag{49}$$

The denominators have physical meaning:

FIG. 7. PMNS mixing angles: UDT predictions (vertical lines) vs. NuFIT 5.2 distributions. All three predictions fall within the 1σ bands.

FIG. 8. The quantum number denominators (13, 7, 45) in the PMNS formulas and their decomposition into the four multiplicities $\{2,3,5,7\}$.

- $13 = 4 + 9 = (2j+1)^2 + (2\ell+1)^2$: spin-orbital partition.
- $7 = 2|\kappa_{\max}| + 1$: closure orbit size.
- $45 = 9\times 5 = (2\ell+1)^2 \times (2|\kappa_{\max}|-1)$: orbital barrier times angular content.

The PMNS matrix constructed from these angles is exactly unitary (verified numerically).

XVI. THE GENERALIZED BRIDGE IDENTITY

The bridge identity (40) is the first row of a three-row structure that transitions between the spin (j) and orbital (ℓ) sectors:

$$\text{source}(\kappa{=}{-n})^2 \times E_1(\kappa{=}{-n})^2 = B(n)\,. \tag{50}$$

Table VIII shows the transition: at $|\kappa| = 1$, the orbital quantum number ℓ governs; at $|\kappa| = 3 = |\kappa_{\max}|$, the spin quantum number j takes over; at $|\kappa| = 2$, both participate and π enters as the S^2 circumference factor.

XVII. CMB PEAK POSITIONS

A. WKB peak law on the sourced profile

The scalar wave equation on the cosmological ϕ profile yields CMB peak positions via the WKB formula:

$$\ell_n = \ell_A\left(\nu_n + \frac{q_1}{\nu_n} + \frac{q_2}{\nu_n^3}\right)\,, \qquad \nu_n = n + c_{\text{eff}}\,, \tag{51}$$

where $\ell_A = 220/F_1$ is the acoustic scale, $c_{\text{eff}} = \arctan(\eta)/\pi$ is the phase offset, and q_1, q_2 are curvature corrections from ϕ derivatives at r_*. All quantities are derived from the metric.

B. Results

With one free parameter (μ_g), the RMS error over seven peaks is 1.32% (Table IX). The Planck data processing pipeline uses ΛCDM transfer functions for foreground subtraction and beam deconvolution, introducing an estimated 2–3% model contamination. A model-independent CMB reduction would be needed for a fairer comparison.

TABLE VIII. Generalized bridge identity: the j–π–ℓ transition across $|\kappa| = 1, 2, 3$.

$\lvert\kappa\rvert$	source$\cdot E_1$	source$^2\cdot E_1^2$	Algebraic	Sector	Match
1	$\sqrt{2}/6$	$1/18$	$1/[2(2\ell+1)^2]$	orbital	0.21%
2	$\pi/6$	$\pi^2/36$	$(\pi/[(2j+1)(2\ell+1)])^2$	mixed	0.11%
3	$\sqrt{2}/4$	$1/8$	$1/[2(2j+1)^2]$	spin	0.045%

FIG. 9. The generalized bridge identity. The product source$^2 \times E_1^2$ transitions from the orbital sector (ℓ-dominated) at $|\kappa| = 1$ through the mixed sector at $|\kappa| = 2$ to the spin sector (j-dominated) at $|\kappa| = 3$.

C. Cross-sector link

The acoustic scale ℓ_A is algebraically related to microphysical eigenvalues:

$$\ell_A = \frac{2\pi r_*(E_2/E_1)}{I_2} = \frac{6\pi r_* \varphi_{\rm gold}}{I_2^2}\,, \tag{52}$$

connecting the CMB peak spacing to the meson mass ratio.

XVIII. JWST PREDICTIONS

In UDT, the universe is static and eternal. There is no Big Bang, no "first epoch" of galaxy formation, and no age–redshift relationship. A galaxy at $z = 12$ is not younger than one at $z = 2$ — it is at a different radial position in the ϕ profile, experiencing different dilation.

This produces a specific prediction:

> *Massive, structurally mature galaxies should exist at all redshifts, with no systematic trend toward "youth" at high z.*

The JWST observations of massive galaxies at $z > 10$ — widely described as "impossibly early" in the ΛCDM framework [11, 12] — are precisely what UDT predicts. The "impossible early galaxies" are simply galaxies seen through a deeper part of the ϕ gradient. In UDT, there is no paradox.

This prediction was inherent in the metric from the start and predates the JWST data. It is qualitative (the galaxy population statistics have not been quantitatively modeled), but the qualitative content — mature galaxies at all redshifts — is confirmed by JWST observations and is in direct tension with ΛCDM expectations.

TABLE IX. CMB peak positions. Best fit with $\mu_g = 0.248$ Gpc^{-1} (1 free parameter). No H_0, Ω_m, Ω_Λ, Ω_b, n_s, or τ.

Peak	$\ell_{\rm pred}$	$\ell_{\rm obs}$ (Planck)	Error
1	220.0	220.0	0.00% (anchored)
2	536.8	537.5	−0.13%
3	831.1	810.8	+2.50%
4	1133.3	1120.9	+1.10%
5	1438.8	1444.2	−0.38%
6	1745.9	1776.0	−1.69%
7	2054.0	2081.0	−1.30%
RMS (peaks 1–7)			1.32%

FIG. 10. CMB angular power spectrum: peak positions. Vertical lines: Planck 2018 measured positions. Points: UDT predictions from WKB on the sourced ϕ profile. RMS: 1.32%.

XIX. ROTATION CURVES

A. First-principles derivation

The geodesic equation on the metric (1) gives the circular velocity:

$$v^2(r) = c^2\, r\, |\phi'(r)|\, e^{-2\phi(r)}\,. \tag{53}$$

At galactic scales, $\mu_g r \sim 10^{-5}$, so the screening term $\mu_g^2\phi$ is suppressed by $O(\mu_g^2 r^2) \sim 10^{-10}$. In the weak-field limit:

$$v^2_{\rm UDT}(r) = \frac{G\, M_{\rm bar}(<r)}{r} = v^2_{\rm Newton}(r)\,. \tag{54}$$

UDT recovers Newtonian gravity exactly at galactic scales. The flat rotation curve problem is open in UDT, as in standard GR.

B. Ancient baryonic remnants

In UDT's static, eternal universe, galaxies accumulate dead stellar remnants (white dwarfs, neutron stars, black holes) over 100–1000 Gyr. Supernova kicks naturally form halos. A one-parameter ($T_{\rm age}$) model applied to 175 SPARC galaxies achieves 96% improvement over baryons-only for massive galaxies, within 1.76× of NFW dark matter halos with half the free parameters [1].

This mechanism requires no new fields, no modified geodesics, and no dark matter — only standard stellar evolution on longer timescales.

XX. BLACK HOLES

A. The positive-ϕ_0 branch

The vacuum classification (Sec. III E) produces black holes from $\phi_0 > 0$. In this branch, $\phi(r) \to +\infty$ at a finite

FIG. 11. SPARC rotation curve fits. Left: UDT baryons-only. Right: UDT with ancient baryonic remnants ($T_{\rm age}$ model). The remnant mechanism closes most of the gap between baryons-only and NFW fits.

radius r_c, where

$$g_{tt} \to 0, \qquad g_{rr} \to +\infty\,, \tag{55}$$

and the Misner–Sharp mass gives $r_c = 2\,m_{\rm MS}$ — the Schwarzschild radius. UDT and GR agree on the horizon location.

B. Differences from Schwarzschild

The UDT black hole (Table X) has no interior singularity. The center is regular ($\phi(0) = \phi_0$), and the "singularity" is at the boundary surface r_c. There is no causal disconnection.

C. Matter dissolution and recycling

A particle cavity ($\phi_0 < 0$) approaching a black hole ($\phi_0 > 0$) experiences a locally increasing ϕ. When the effective ϕ_0 crosses zero, the eigenvalue spectrum ceases to exist (Sec. IX): the particle dissolves. This occurs *outside* the horizon, providing a natural mechanism for matter recycling in a static universe.

XXI. PREDICTIONS AND FALSIFICATION

A theory with 30+ predictions must specify the conditions under which it would be falsified. Table XI lists concrete falsification criteria with named experiments.

The most immediate tests:

1. **JUNO** (2025–2030): will measure $\sin^2\theta_{12}$ to ± 0.005, testing the prediction $4/13 = 0.3077$ at $> 3\sigma$ sensitivity.

2. **KATRIN/Project 8**: direct neutrino mass measurements approaching the 0.04–0.06 eV range where $m_\nu = \alpha^3 m_e/4 = 0.049$ eV lies.

TABLE X. Structural differences between UDT and Schwarzschild black holes.

Property	Schwarzschild	UDT
Horizon	$g_{tt} = 0$ at $r = 2M$	$g_{tt} \to 0$ asymptotically
Interior	Sign swap	No sign swap
Singularity	At $r = 0$	At $r = r_c$ (boundary)
Center	Singular	Regular (ϕ_0 finite)

FIG. 12. Falsification landscape. Each point represents a prediction; the horizontal axis shows current experimental precision; the vertical axis shows the UDT prediction. Arrows indicate experiments that will improve precision into the falsification range within 5 years.

3. **Sandage–Loeb test** (ELT drift spectroscopy): will directly test cosmic expansion vs. static geometry over 10–20 year baselines.

4. **JWST/Roman**: continued high-z surveys will either find or rule out massive mature galaxies at $z > 15$.

XXII. DISCUSSION

A. Summary of the framework

The results presented in this paper flow from a single line element [Eq. (1)] through a chain of standard mathematical operations: tetrad construction, spin connection, covariant Dirac equation, Sturm–Liouville eigenvalue theory, angular coupling integrals on S^2, and the WKB approximation for the scalar wave equation. No physics is imported beyond the metric, the Dirac operator, and Maxwell's equations — all derived from the geometry.

The three parameters (ϕ_0, μ^2, r_*) are determined by angular quantum numbers on S^2, and the single calibration C sets the mass scale via the electron mass. From this minimal input, the theory produces:

- Parameter-free lepton mass ratios accurate to 0.002–0.03%.
- The pion mass from the Diophantine identity $84 = \binom{9}{3}$ at 0.09%.
- A meson–baryon spectrum matching 17 particles within 1%.
- The fine-structure constant at 0.29%.
- Three PMNS mixing angles, all within 1σ.
- The atmospheric neutrino mass at 0.5%.
- SNe distances at 0.166 mag RMS.
- CMB peak positions at 1.32% RMS.
- BAO distances at 3.8% RMS.

TABLE XI. Falsification criteria. Each row identifies a prediction, the threshold for falsification, and the experiment or measurement that would test it.

Prediction	Falsification criterion	Experiment/Measurement
$m_p/m_e = 6\pi^5$	Ratio deviates $> 0.01\%$ from $6\pi^5$	CODATA precision mass measurements
$m_\mu/m_e = 20\pi^3/3$	Ratio deviates $> 0.1\%$	Muon $g-2$ experiments (Fermilab, J-PARC)
$m_\pi = 84\pi\, m_e$	Ratio deviates $> 0.5\%$	PIENU, precision pion mass
$1/\alpha_{\rm EM} = 36\pi/I_2$	Deviation $> 1\%$	Electron $g-2$ (Harvard)
$\sin^2\theta_{12} = 4/13$	Deviation $> 3\sigma$	JUNO, DUNE
$\sin^2\theta_{23} = 4/7$	Deviation $> 3\sigma$	Hyper-Kamiokande, DUNE
$\sin^2\theta_{13} = 1/45$	Deviation $> 3\sigma$	JUNO
$m_\nu = \alpha^3 m_e/4$	Direct mass > 0.06 eV or < 0.04 eV	KATRIN, Project 8
$m_n - m_p = (5/4)\alpha_{\rm EM} C$	Deviation $> 2\%$	Lattice QCD independent verification
Static universe	Detection of cosmological expansion at $> 5\sigma$	Sandage–Loeb test (ELT)
Mature galaxies at all z	Systematic absence of mature galaxies at $z > 15$	JWST, Roman Space Telescope
No dark matter particle	Detection of WIMP or axion	LZ, XENONnT, ADMX
$c^2 = 2GM/r_*$	Machian relation fails at $> 1\%$	Improved BAO + CMB joint analysis

B. The static universe

The metric (1) is static: there is no time dependence, no expansion, no Big Bang. Cosmological redshift arises from the spatial gradient of ϕ [Eq. (5)], not from the expansion of space. The CMB is the last-scattering surface at the radial distance where $\phi(r_*) = \ln(1 + z_{\rm CMB}) = 7.004$.

This is a legitimate general-relativistic solution. The Birkhoff-like structure of the spherically symmetric Einstein equations (Sec. III) admits both static and dynamic solutions; we work with the static branch. Whether the universe is expanding or static is an observational question, and we note that the Sandage–Loeb test [9, 10] — a model-independent measurement of cosmic drift — has not yet been performed at the required sensitivity.

C. The role of ϕ

The scalar field $\phi(r)$ plays a dual role:

1. **Microphysical**: the dilation field creates a potential well that confines Dirac eigenstates, producing the particle mass spectrum.

2. **Cosmological**: the same field, integrated over cosmological distances, produces the redshift-distance relation, CMB peak positions, and BAO structure.

The two regimes are cleanly separated: the microphysical cavity has $r_* \approx 7$ in natural units (~ 7 fm), while the cosmological boundary has $r_* \approx 10$ Gpc. The same equation [Eq. (7)] governs both, with different parameters (μ vs. μ_g) and different sources.

D. What this theory does not explain

We note the following open problems:

- **Rotation curves**: UDT recovers Newtonian gravity at galactic scales. The flat rotation curve problem is open, though the ancient baryonic remnant mechanism is promising.

- **CMB peak heights**: the full power spectrum (including peak amplitudes) is not yet competitive with ΛCDM.

- **The tau lepton**: no mass formula has been derived.

- **The CP-violating phase**: the fourth PMNS parameter δ is not yet predicted.

- **Cross-sector closure**: the mechanism connecting microphysical and cosmological parameters is identified (shared geometric constants) but not fully derived from a single action.

E. Comparison with ΛCDM

The standard cosmological model uses six parameters (H_0, Ω_b, Ω_m, n_s, A_s, τ) to fit cosmological data. UDT uses one cosmological parameter (μ_g) and achieves comparable results on peak positions, while falling short on peak heights and the full CMB power spectrum.

On particle physics, ΛCDM has no predictions (particle masses are free parameters in the Standard Model). UDT derives 17+ particle masses and three mixing angles from the metric — content that the Standard Model takes as input.

The comparison is not symmetric: UDT attempts to unify particle physics and cosmology from a single geometric object, while ΛCDM addresses only cosmology and treats particle physics as external input.

XXIII. CONCLUSION

We have shown that the static, spherically symmetric metric

$$ds^2 = -e^{-2\phi(r)}\, c^2\, dt^2 + e^{2\phi(r)}\, dr^2 + r^2\, d\Omega^2$$

with a single screened scalar field $\phi(r)$ produces more than 30 quantitative predictions spanning particle masses, coupling constants, neutrino parameters, and cosmological observables. All predictions flow from the metric through standard mathematical operations, with three geometric parameters and one mass anchor.

The key results — $m_p/m_e = 6\pi^5$ at 0.002%, $m_\pi = 84\pi\, m_e$ at 0.09%, $1/\alpha = 36\pi/I_2$ at 0.29%, three PMNS angles within 1σ, and the atmospheric neutrino mass at 0.5% — are either correct or very precisely wrong. The falsification criteria (Table XI) specify the experiments that will distinguish these two possibilities within the next decade.

This is a modest extension of general relativity: one metric, one scalar field, one principle (everything from the geometry). The metric was published in September 2025. All numerical codes, derivations, and datasets are available at [1].

ACKNOWLEDGMENTS

The authors acknowledge the use of the Pantheon+ supernova compilation [4], Planck 2018 CMB data [3], BOSS and DESI BAO measurements [5], SPARC rotation curve data [7], the Particle Data Group compilations [2], and the NuFIT neutrino oscillation global fit [6]. Numerical computations were performed on a Tesla V100 GPU using PyTorch for vectorized ODE integration. This work was supported by no external funding. All code is available under CC BY-NC-ND 4.0 at [1].

Appendix A: The Einstein Tensor

For the metric (2), the nonzero mixed components of the Einstein tensor are:

$$G^t{}_t = -\frac{1}{r^2}\left[1 - e^{-2\phi}(1 - 2r\phi')\right], \tag{A1}$$

$$G^\theta{}_\theta = e^{-2\phi}\left[-\phi'' + 2(\phi')^2 - \frac{2\phi'}{r}\right], \tag{A2}$$

with $G^r{}_r = G^t{}_t$ (identically) and $G^\varphi{}_\varphi = G^\theta{}_\theta$.

The Bianchi identity $\nabla_\mu G^{\mu\nu} = 0$ gives the integral form

$$G^t{}_t = \frac{1}{r^2}\left[C_0 + I(r)\right], \tag{A3}$$

where $C_0 = e^{-2\phi_0} - 1$ and $I(r) = -2\mu^2 \int_0^r s\,\phi(s)\, ds$. Both decompositions are verified numerically to residual $< 2\times 10^{-9}$.

Appendix B: Misner–Sharp Mass and the Machian Relation

The Misner–Sharp quasi-local mass for the metric is

$$m(r) = \frac{c^2\, r}{2G}\left(1 - e^{-2\phi(r)}\right). \tag{B1}$$

At the boundary $r = r_*$ with total mass M:

$$c^2 = \frac{2GM}{r_*(1 - e^{-2\phi(r_*)})}. \tag{B2}$$

In the deep-well limit $|\phi(r_*)| \gg 1$ ($e^{-2\phi_*} \to 0$):

$$c^2 \approx \frac{2GM}{r_*}, \tag{B3}$$

verified to $< 0.01\%$ at BAO-constrained parameters. The cosmological boundary value $\phi(r_*) = \ln(1 + z_{\rm CMB}) = 7.004$ is algebraically fixed by the combination of the redshift definition (5) and the Misner–Sharp relation (B1).

Appendix C: Dirac Stress-Energy Tensor

For a massless Dirac eigenstate (G, F, E, κ) on the UDT metric, the stress-energy components (after angular integration) are:

$$T^t{}_t(r) = -\frac{E\, e^{\phi}(G^2 + F^2)}{4\pi\, r^2}, \tag{C1}$$

$$T^r{}_r(r) = T^t{}_t + \frac{2\kappa\, e^{-\phi}\, G\, F}{4\pi\, r^3}, \tag{C2}$$

$$T^\theta{}_\theta(r) = -\frac{1}{2}(T^t{}_t + T^r{}_r). \tag{C3}$$

The trace vanishes exactly: $T^\mu{}_\mu = 0$ (for $m = 0$). The ratio $R \equiv T^r{}_r/T^t{}_t$ ranges from 0.094 (ground state) to 0.841 ($n = 3$ excited), approaching 1 in the ultrarelativistic limit. The constraint $G^t{}_t = G^r{}_r$ requires $R = 1$ for total matter, which is achieved by including the electromagnetic sector ($R_{\rm EM} = 1$ exactly).

Appendix D: The Vacuum Classification Theorem

Define $y(r) := e^{-2\phi(r)} > 0$. The vacuum flux system is equivalent to

$$y'' + \frac{2}{r}\, y' = \mu^2 \ln y, \quad y(0) = e^{-2\phi_0}, \quad y'(0) = 0. \tag{D1}$$

Theorem. For $\mu > 0$ and regular origin data:

(i) $\phi_0 > 0$: y reaches zero at finite r_c ($\phi \to +\infty$, black hole horizon).

(ii) $\phi_0 = 0$: $y \equiv 1$, $\phi \equiv 0$ (trivial).

(iii) $\phi_0 < 0$: y is globally defined, strictly increasing, $y \to \infty$ ($\phi \to -\infty$).

Proof. Case (i): $y_0 < 1$, so $\ln y < 0$, driving y downward. The quadratic bound $y(r) \leq y_0 - (\mu^2 a/6)\, r^2$ with $a = -\ln y_0 > 0$ guarantees $y = 0$ at finite r_c. Case (iii): $y_0 > 1$, so $\ln y > 0$, driving y upward. Global existence follows from $y(r) \leq y_0 \exp(\mu^2 r^2/6)$. A finite limit $L > 1$ leads to $y' \sim cr$ for large r, contradicting convergence. □

Appendix E: The Separatrix Theorem

For the sourced scalar equation with ε-decomposition $S(r) = \mu^2\phi(1-\varepsilon)$, the *null-source curve* satisfies $J' = 0$ everywhere. Setting $u = \phi'(r)$ gives a Bernoulli ODE with solution:

$$\phi'_{\rm sep}(r) = \frac{1}{r(2+Cr)}\,, \tag{E1}$$

where C is an integration constant (distinct from the mass calibration). Any profile with ϕ' above this curve is in the drive regime; below is in the brake regime. The sign-change radius r_{sc} (maximum of the scalar flux) satisfies $J'(r_{sc}) = 0$ and is determined entirely by the profile coefficients, with no external input.

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